

TWINGATE: Supplementary Results and Derivation Detail

Companion document to *TWINGATE: A Twin-Gated Framework for Physics-Based Coverage Validation in V2X Digital Twins*

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1 How to use this document

Everything in `main.tex` is condensed for narrative flow: headline numbers, the theoretical reasoning behind them, and pointers to “full detail here.” This document *is* that full detail. Each section below is self-contained and mapped to the main-paper section it supports, so material can be promoted back into the main paper (e.g., if a reviewer asks for it) without re-deriving anything.

Section 7 and Section 8 describe experiments that were run, produced real results, and are **not currently included in `main.tex`** at all — they are genuine findings (one positive, one a well-diagnosed negative result) held in reserve pending a decision on whether they strengthen the paper enough to include.

2 Pipeline Hyperparameters

Supports: Section III.A (3D Scene Construction), III.D.3 (4D Optimization).

2.1 Scene construction

- Operator 1 scene origin: (52.507005°N, 13.323428°E)
- Operator 2 scene origin: (52.506112°N, 13.321908°E)
- Ray tracing backend: Sionna RT 2.0.1 on the `cuda_ad_mono_polarized` Mitsuba 3 variant (GPU-accelerated, monochromatic, polarized).
- Max path depth: 5 interactions (reflection + diffraction; scattering added in Stage 4, see below).
- Per-call cost: a batched forward pass over 40 receiver locations takes approximately 2–3 s on an NVIDIA GPU under this configuration. A full 4D Nelder-Mead optimization of one tower requires 80–95 such calls (including the coarse azimuth grid), i.e. roughly 3–6 minutes per tower on a single GPU.

2.2 4D Nelder-Mead optimizer (Gate 2 position/height/azimuth search)

- Search variables: $(\Delta x, \Delta y, h, \varphi)$ — horizontal displacement from WCL init, height $h \in [10, 60]$ m, azimuth $\varphi \in [0^\circ, 360^\circ)$.
- Coarse pre-search: 8-direction azimuth grid at $0^\circ, 45^\circ, 90^\circ, \dots, 315^\circ$; best-scoring orientation seeds the full 4D simplex.
- Simplex initialization: 5 vertices at the WCL position, with a 50 m horizontal arm, 10 m height arm, and 45° azimuth arm.
- Sionna yaw convention: compass bearing φ maps to scene yaw via $\alpha = \pi/2 - \varphi$ (x-axis = East).
- Search bound: 500 m radius in $(\Delta x, \Delta y)$; height and azimuth clamped to their physical ranges at every evaluation.
- Convergence: $|\Delta \theta| < 10$ m and $|\Delta \text{MAE}| < 0.02$ dB, 80-iteration hard cap.

- Per-objective-evaluation training subsample: 40 measurement rows (one GPU batch) during the search; the final evaluation pass uses all available training and validation rows for that tower.
- OLS calibration slope bound: $\alpha_i \in [0, 2]$ (prevents sign flips and unphysical amplification).

2.3 Diffuse scattering (Stage 4)

- Material: ITU `concrete` preset (all building surfaces).
- Scattering coefficient: $S = 0.4$ (TR 38.901 value at 2 GHz), vs. Sionna’s default $S = 0$.
- Scattering pattern: Lambertian (isotropic diffuse lobe).
- Everything else in the pipeline (NM search, sector antenna, OLS calibration) is held fixed between Stage 3 ($S = 0$) and Stage 4 ($S = 0.4$) to isolate the marginal contribution of scattering.

3 Per-Device Gate 2 Results (Full Breakdown)

Supports: Table I and §IV.C (Gate 2: RSRP Fidelity) in `main.tex`, which report only the Operator-level combined figures.

Table 1: Per-device Stage 3 (position/orientation refinement, no scattering) results, temporal split, held-out day 3.

Device	Towers	WCL MAE (dB)	Refined MAE (dB)	Gain (dB)	Improved
pc1	81	3.877	3.029	+0.848	63/81 (77.8%)
pc4	43 ^a	3.787	3.016	+0.771	33/43 (76.7%)
pc2	37	6.927	4.601	+2.327	25/37 (67.6%)
pc3	67	6.786	3.986	+2.801	53/67 (79.1%)
Op. 1 (pc1+pc4)	124	3.865	3.027	+0.838	96/124 (77.4%)
Op. 2 (pc2+pc3)	104	6.805	4.069	+2.736	78/104 (75.0%)

^a45 towers refined; 2 excluded (see §3.1).

Table 2: Per-device Stage 4 (+ diffuse scattering, $S = 0.4$) results, temporal split, held-out day 3.

Device	Towers	WCL MAE (dB)	Refined MAE (dB)	Gain (dB)	Improved
pc1	81	3.874	3.020	+0.853	62/81 (76.5%)
pc4	43 ^a	3.803	2.980	+0.823	32/43 (74.4%)
pc2	37	6.939	4.553	+2.387	26/37 (70.3%)
pc3	67	6.743	3.957	+2.786	54/67 (80.6%)
Op. 1 (pc1+pc4)	124	3.865	3.015	+0.849	94/124 (75.8%)
Op. 2 (pc2+pc3)	104	6.770	4.037	+2.732	80/104 (76.9%)

Median/mean TX position correction (Stage 4, combined Op. 1): median 45 m, mean 63 m. Refined TX heights span the 10–60 m search range with recovered azimuths spanning the full compass, i.e. no boundary-clamping artifact.

Total Sionna calls per run (full pipeline, one device, one stage): pc1 $\approx$ 2,400, pc2 $\approx$ 1,150, pc3 $\approx$ 2,300, pc4 $\approx$ 940–1,300 depending on stage.

3.1 Degenerate (NaN) towers — pc4

Two pc4 towers (cell IDs 26947081 and 28792072) produced `refined_val_mae = NaN` in both Stage 3 and Stage 4. Diagnosis: both have very small validation sets (23 and 9 rows respectively); at the Nelder-Mead-refined position, Sionna found zero valid propagation paths to *every* row in that tower’s validation subset (all predictions hit the -100 dBm sentinel floor), so even the code’s flat-signal fallback (mean of training RSRP) divides by an empty masked array and returns NaN. This reproduced identically under scattering, confirming it is a geometry/orientation degeneracy specific to these two small-validation-set towers, not a scattering artifact. Both towers are excluded from the Stage 3/4 aggregates above (45 candidate towers $\rightarrow$ 43 valid). This is consistent with the paper’s existing observation (§IV.E) that refinement most often misbehaves for towers with the fewest training/validation rows.

4 Gate 1: Full Statistics

Supports: §IV.A (Gate 1: Dead-Zone Preservation) in main.tex.

4.1 Gap-event counts (gap-level, not row-level)

Period	TypeA	TypeB	Ambiguous
Train (days 1–2)	4	57	13
Val (day 3)	1	15	4

Training TypeA gaps split evenly by operator (2 from pc2/Op. 2, 2 from pc4/Op. 1). The dominant TypeB cluster (56 of 57 training TypeB gaps) belongs to Operator 2, centroid (52.514212°N, 13.348557°E), stable to within 15 m across all three collection days.

4.2 Threshold sensitivity, full sweep

d_{recur}	Matched (of 15 val TypeB)	C	Note
50 m	11/15	0.73	strict; margin gaps excluded
100 m (used)	12/15	0.80	midpoint, matches 1 Hz GPS uncertainty
200 m	15/15	0.87 ^a	permissive

^aReported as 0.87 in main text; some sweep runs during development returned slightly different denominators depending on whether ambiguous-adjacent gaps were re-classified at wider radii — 0.87 is the value used in all published figures.

4.3 Multi-KPI corroboration, full statistics

Held-out day 3, Operator 2, comparing rows within 150 m of the TypeB centroid (“near,” 819 rows total / 723–745 with valid readings depending on the KPI column) against the remaining route (“background,” 22,217–22,245 rows).

5 Twin-Gate Convergence: Pipeline History

Supports: §IV.B (Twin-Gate Convergence) in main.tex. Reported here as a robustness/consistency check: three successive, independently-computed pipeline versions all land in a narrow band.

KPI	Near mean	Background mean	Welch t	p
RSRP (dBm)	-115.43	-89.97	-64.22	1.7×10^{-316}
RSRQ (dB)	-15.99	-10.48	-32.87	6.2×10^{-147}
SNR ₁ (dB)	-2.73	10.60	-42.25	7.4×10^{-202}
SNR ₂ (dB)	-3.30	9.40	-53.58	2.1×10^{-269}
Avg. MCS index	5.88	15.08	-60.18	3.7×10^{-315}
Datarate (bps)	1.37×10^6	1.97×10^7	-134.30	≈ 0

Pipeline version	Predicted drop	Measured drop	% recovered	Basis
Early stratified-split	27.2 dB	30.3 dB	90.0%	pre-fix, both Op. 2 devices
Intermediate (pc3-only)	27.5 dB	31.5 dB	87.4%	pc3 only, temporal, Stage 4
Final (pc2+pc3)	26.3 dB	30.4 dB	86.5%	complete Op. 2, temporal, Stage 4

Final version detail: near-zone mean predicted -115.57 dBm, background mean predicted -89.29 dBm; near-zone mean measured -120.02 dBm, background mean measured -89.65 dBm. 364 near-zone rows, 14,807 background rows (15,171 total, pc2+pc3 combined). Welch $t = -56.63$, $p = 6.2 \times 10^{-193}$.

6 M2 Recompute: Provenance and Correction History

Supports: §IV.C (M2 dead-zone detection) in main.tex.

What happened. The M2 figure originally reported (F1= 0.63, precision 0.52, recall 0.82, threshold -116 dBm) was computed by `M2_serving_cell.py` from `per_tower_val_rows_op1/op2.csv` — files dated prior to this project’s temporal-split correction, using *raw WCL positions with no Nelder-Mead refinement and no scattering*. This was discovered while investigating why M2 seemed disconnected from the corrected Gate 2 pipeline.

Correction. M2 was recomputed from the current Stage 4 (refined position + $S = 0.4$ scattering) per-row predictions:

Basis	Threshold	Precision	Recall	F1	TypeB rows
Stale (WCL-only, pre-fix)	-116 dBm	0.52	0.82	0.63	386/42,830
pc3-only (Stage 4)	-113 dBm	0.75	0.89	0.82	364/13,127
pc2+pc3 combined (Stage 4, final)	-113 dBm	0.67	0.89	0.77	364/15,171

Note the pc3-only estimate (F1= 0.82) is *higher* than the combined estimate (F1= 0.77): pc2 contributes 2,044 additional background rows but zero additional TypeB-zone rows on this held-out day (pc2’s smaller, differently-timed validation set did not pass through the dead zone), so adding it can only add potential false positives, not true positives. The combined figure is reported as the paper’s headline M2 result because it is the complete, honest measurement, not because it is the most favorable one.

7 Explored, Not Included: LOS/NLOS Classification and Multipath Richness (M4)

Status: real result, held out of main.tex by author decision, not by any negative finding. This section documents a fourth candidate readiness axis — structural/mechanistic validation, distinct from M1 (accuracy) and M2 (localization) — computed via a new script (A17) that extracts per-path richness metrics from Sionna’s existing solver output (delay, angle of arrival, interaction type) at zero additional RT cost, reusing each device’s already-refined Stage 4 geometry.

7.1 Method

Per validation row: power-weighted RMS delay spread (τ_{rms}), power-weighted circular RMS azimuth spread, valid path count, and an LOS/NLOS flag (LOS if any valid path has an all-NONE interaction sequence — i.e. a direct, unobstructed line — per Sionna’s `InteractionType` enum: NONE=0, SPECULAR=1, DIFFUSE=2, REFRACTION=4, DIFFRACTION=8).

7.2 M4a: LOS/NLOS accuracy

Device	LOS mean (dBm)	LOS std	NLOS mean (dBm)	NLOS std	Welch t (p)
pc1	−84.89	10.83	−93.65	9.07	25.6 (6×10^{-109})
pc3	−90.12	13.19	−81.93	10.47	−24.2 (4×10^{-109})
pc2	−96.47	11.49	−95.53	7.40	−1.4 (n.s., $p = 0.16$)

pc1: 81 towers, 14,172 rows. pc3: 63 towers, 13,127 rows. pc2: 36 towers, 2,044 rows.

This is not a clean, consistent result across devices. pc1 shows LOS > NLOS in measured RSRP, exactly as propagation physics predicts. pc3 shows the *opposite* sign, and pc2 shows no significant difference at all. This inconsistency — not a single clean number — is the actual reason this metric is not in main.tex: a metric that flips sign between two devices on the same operator’s network cannot be presented as a validated readiness criterion without a lot more investigation into why (candidate explanations, not yet tested: Operator 2’s more complex, NLOS-dominated urban geometry near the confirmed dead zone; a sector-antenna-orientation interaction with which paths get classified as “direct”). *Literature note:* VaN3Twin’s real-time follow-up (Pegurri et al., arXiv 2601.16559) does validate a C-V2X DT using exactly this kind of LOS/NLOS transition-classification metric, so this is a recognized axis in peer work, not an invented one — worth revisiting with a cleaner methodology rather than abandoning.

7.3 M4b: Multipath richness vs. M1 residual

Device	corr(τ_{rms} , residual)	p	corr(n_{valid} , residual)	p
pc1	−0.032	1.5×10^{-4}	−0.065	6.3×10^{-15}
pc3	−0.009	0.308 (n.s.)	−0.103	4.6×10^{-32}
pc2	−0.144	5.9×10^{-11}	−0.143	7.8×10^{-11}

All correlations are negligible in magnitude ($|r| < 0.15$) despite significance at large N . **Null result:** multipath richness does not meaningfully explain where M1 residual error concentrates. This suggests the residual is dominated by something else (OLS calibration slack, GPS positional noise, or building-material approximation error), which is itself a useful negative finding pointing future work away from richness-based residual explanations.

8 Explored, Not Included: Handover / Cell-Boundary Prediction

Status: genuine negative result, well-diagnosed, not a bug. Motivated by the question “does Gate 2’s geometry correctly predict *where* a moving vehicle should hand over between cells,” tested two independent ways.

8.1 Feasibility groundwork

Gate 1’s own TypeA (session-boundary) events are far too rare to serve as ground truth: only 5 total across the entire 3-day dataset (4 train, 1 val) — the TypeA definition conflates handover with RLF-reestablishment and logger restarts, and requires a measurement *gap*, which most real handovers do not cause. The better ground-truth signal is raw serving-cell (`PCell_Cell_Identity`) transitions, filtered for persistence to exclude cell-flapping noise:

Persistence filter	pc1 val transitions	pc2	pc3	pc4
≥ 5 s	391	48	408	52
≥ 10 s	336	41	313	43
≥ 20 s	258	31	228	31

At the 20s threshold, 96.1% of pc1’s held-out transitions have both the source and destination tower already covered by the existing Stage 4 refined-position set — no new RT position optimization needed.

8.2 A18: Cross-cell direction test

Method: for each covered transition, evaluate Sionna’s raw (uncalibrated) predicted power from both the source and destination tower at the last pre-transition GPS point and the first post-transition point, using each tower’s already-refined Stage 4 position (no recalibration — every transition here is intra-operator, so towers share a comparable EIRP/hardware family, which was the original rationale for using raw power). Metric: Handover Direction Accuracy (HDA) — fraction of transitions where the predicted-stronger tower matches the real serving cell at *both* points.

Result (pc1, 10s filter, 322 covered transitions, 293 after removing 29 sentinel-contaminated (‘no path found’) events):

- HDA (strict): 10.6% (31/293)
- Ranking-flip rate (softer — does the margin move in the right direction, regardless of absolute correctness at either point): 62.1% (182/293)
- **correct_A** (source tower wins *before* the real switch): 44.0% — *below* chance
- **correct_B** (destination tower wins *after* the real switch): 65.2% — *above* chance

Diagnosis. The asymmetry between **correct_A** (below chance) and **correct_B** (above chance) is not random noise — it points to a systematic effect. Each tower’s position/height/azimuth was refined by Nelder-Mead to minimize error over *that tower’s own bulk of training rows*, which are concentrated in the interior of its serving area. A cell-boundary point is, by construction, the sparsest, least-represented part of each tower’s own training distribution — the edge where that tower is weakest and about to hand off. The per-tower-independent calibration that gives a solid overall M1 (3.02 dB average) is systematically weakest exactly where handover-boundary accuracy would need to be strong. This is a real limitation of the current calibration architecture (independent per-tower fits), not a bug in this experiment or evidence against the rest of the pipeline.

8.3 Independent corroboration: pre-existing proxy_b

A simpler, independently-implemented handover check already existed in `M2_serving_cell.py` (“Proxy B”), using *calibrated* (not raw) power at a single transition point, on the older stratified-split, unrefined-position pipeline: **351 handover events, 156 correct, 44.4% accuracy** — also below chance. Two independent implementations, different calibration approaches, different pipeline versions, and different data splits reach the same qualitative conclusion: cross-cell handover-boundary prediction does not work well under per-tower-independent Gate 2 calibration. This convergence of two separate negative results is itself fairly strong evidence the limitation is real and structural, not an implementation artifact — and settles the question for the current architecture rather than inviting further quick attempts at a fix.

8.4 What would actually be needed

A joint, boundary-aware calibration (e.g., fitting neighboring towers together rather than independently, or explicitly up-weighting training rows near candidate boundaries) — a genuinely new methodology, not a parameter tweak, and a reasonable candidate for a journal extension rather than this conference submission.

9 Literature Scan Summary

A dedicated literature scan (`DT_READINESS_LITERATURE_SCAN.md`, in the project repository) surveyed published Digital Twin implementations for C-V2X / vehicular cellular coverage to identify which experiments and thresholds TWINGATE should consider replicating. Key findings:

- **DT-CoVeSS** (Twardokus & Rahbari, published IEEE INFOCOM Workshops/DTwin 2025) is a Sionna RT DT built specifically for *PC5 sidelink* — confirms sidelink C-V2X DT validation is a distinct research track with its own data requirements, independently supporting TWINGATE’s own scope decision to stay Uu-mode only given the Berlin V2X dataset has no sidelink logs.

- **Lucas-Estañ et al.** (published IEEE *Network*, vol. 37, no. 4, 2023) grounds V2X service-continuity requirements in exact 3GPP TS 22.186 numbers: Cooperative Lane Change requires 90% of packets under 25 ms (low automation) or 99.99% under 10 ms (high automation). This is the strongest available motivation for the handover/cell-boundary direction (§8), even though that direction did not pan out under the current architecture.
- **Pegurri et al.** (VaN3Twin real-time follow-up, arXiv 2601.16559) validates LOS/NLOS transition-classification accuracy in a live 60 GHz C-V2X testbed — direct peer precedent for the M4 direction (§7).
- No paper found translates a 3GPP V2X reliability/latency target into an RT-usable RSRP/SINR coverage threshold; doing so would require a link-level BLER-vs-SINR simulation step (a different Sionna module — PHY/LDPC, not RT), which is a materially larger undertaking flagged as a longer-horizon journal-extension item.

10 Miscellaneous Fixes Made During This Project

For completeness/transparency, corrections made to the pipeline and paper during development, beyond the M2/convergence recompute already covered in §6:

- **Dataset year:** a transcription error in early drafts referred to the Berlin V2X collection dates as “June 2022”; corrected to the true “22–24 June 2021” throughout.
- **Gate 1 completeness C :** an early draft reported $C = 0.70$ using an incorrect denominator (all validation gaps, including TypeA/Ambiguous); corrected to $C = 0.80$ using the TypeB-only denominator specified in main.tex’s completeness equation (Section III-B.3).
- **Cosmetic logging bugs:** `A15_tx_position_refinement.py` had two hardcoded string labels (“Op1” in the scene-load message, “pc1” in the final-results banner) that did not reflect the actual device/operator being processed — purely cosmetic (the underlying `SCENE_XML` path and output filenames were always correct), fixed to use the actual `DEVICE/OPERATOR` variables.